



FRANTIŠEK RŮŽIČKA

physicist by heart, game dev by occupation

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Prague, Czechia

TECHSTACK

Languages:

C++

C#

Python

Engines/Frameworks:

UE4/5

Unity

Libs/plugins:

Bullet (physics)

PhysX (physics)

Photon Fusion (MP)

Nakama (MP)

UE GAS

LONG-TERM PROJECTS

Unity developer @ Shiny Dynamics

Dec 2024 - Now

remote

- working on a kart racing game prototype with jetskis and water physics
- established the physics, gameplay, audio and netcode mechanics from scratch

C++ Engine developer @ Six Impossible Things Ltd.

Sep 2023 - Nov 2025

remote

- creating physics for a next-gen game engine from scratch
- incl. rapid soft body clothing & fluid simulations, multicore & network friendly

Unreal Engine (C++) Gameplay developer @ Star Atlas

Aug 2020 - Sep 2023

remote

- designed and implemented physics-based spaceship and hoverbike flight models
- general spaceship gameplay (shooting, mounting, collision, destruction) with GAS
- ship & character configuration and customization using GAS

C++ Physics developer @ SCS software

Jul 2016 - Jul 2020

Prague, Czechia

- in charge of a core truck driving model overhaul
- designed & implemented simulated colliding cables, flags, trailer curtains
- transitioned the company's in-house engine from Bullet to PhysX.

SHORT-TERM PROJECTS

Unreal Engine (C++) dev @ Tiny Bull

6 months in 2022

remote

- made an Unreal plugin for a new experimental finger-based game controller

Unreal Engine (C++) dev @ 9axis Ltd.

8 months in 2021

remote

- made a set of 4 physics-based prototypes for a new multi axis controller.
- this includes skateboarding, drone flight, ice hockey and space battle

EDUCATION

MSc. in quantum physics @ Czech Technical University

Sep 2010 - Jun 2015

Prague, Czechia

- graduated with honors in theoretical quantum physics
- 12 articles in impacted journals, h-index 8